

Final Project Report — CineScope Movie Catalog System

1. Introduction

The purpose of this project was to design and develop a modern, user-friendly movie catalog and review platform using **ASP.NET Core MVC**.

The system allows users to browse movies, view details, explore image galleries, and read reviews, while administrators can manage movies, genres, and media assets.

The project integrates external APIs, supports manual media uploads, and provides a clean, responsive UI suitable for both academic evaluation and real-world portfolio use.

2. Project Objectives

The main objectives of the CineScope system were:

- Build a complete **ASP.NET MVC application** with full CRUD functionality
- Implement **TMDB API integration** for automatic poster and image retrieval
- Support **manual image uploads** for custom content
- Provide a clean **admin dashboard** for managing movies and genres
- Display **movie details, reviews, and ratings**
- Create a modern, responsive UI using **Bootstrap 5**
- Document the system professionally for academic submission

All objectives were successfully achieved.

3. System Architecture

The CineScope system follows the **MVC (Model-View-Controller)** architecture:

✓ Models

- Movie
- Genre
- Review
- Rating

✓ Controllers

- MoviesController
- GenresController

✓ Services

- `TmdbService` for external API communication

✓ Views

- Razor views for Movies, Genres, Shared layouts, and partials

✓ Static Assets

- Posters stored in `wwwroot/posters/`
- CSS, JS, and Bootstrap assets in `wwwroot`

The architecture is modular, scalable, and easy to maintain.

4. TMDB API Integration

A major feature of the project is the integration with **The Movie Database (TMDB)**.

The system supports:

- Automatic poster fetching
- Searching TMDB by movie title
- Retrieving movie IDs
- Fetching additional images (posters + backdrops)
- Displaying images in a gallery on the Details page

This makes the application dynamic and visually rich without requiring manual data entry.

5. Manual Image Upload System

To support custom movies and local content, a **manual poster upload** feature was implemented.

Key features:

- Upload images directly from the computer
- Save files to `wwwroot/posters/`
- Automatically generate unique filenames
- Store the file path in `PosterUrl`

- Coexist with TMDB auto-fetch logic

This gives administrators full control over movie artwork.

6. Movie Management (CRUD)

The MoviesController was fully rebuilt to support:

- Create
- Edit
- Delete
- Details
- Index listing
- Genre selection
- Poster upload
- TMDB auto-fetch
- Extra image gallery

All CRUD operations were tested and verified.

7. User Interface Enhancements

The UI was improved to provide a clean, modern experience:

- Bootstrap 5 layout
- Responsive movie cards
- Updated Create/Edit forms
- Poster upload field
- TMDB auto-fetch hint
- Image gallery on Details page
- Clear admin vs. user separation

The interface is intuitive and visually appealing.

8. Testing & Validation

The following features were tested:

✓ **TMDB poster auto-fetch**

Works for Create and Edit when PosterUrl is empty.

✓ Manual poster upload

Files save correctly and display on the site.

✓ Extra images

TMDB backdrops and posters load in the gallery.

✓ CRUD operations

All admin actions function correctly.

✓ Routing & navigation

All pages load without errors.

✓ Sample movies

10 placeholder movies were added for testing.

The system is stable and production-ready.



9. Installation & Setup

A full installation guide was added to the README, including:

- Cloning the repository
- Restoring dependencies
- Adding TMDB API key
- Running migrations
- Starting the application
- Creating an admin user

This ensures anyone can run the project easily.



10. Documentation

The project includes:

- README.md (professional, complete)
- Installation steps
- GIF demo placeholder
- Project summary
- Final project report (this document)

The documentation meets academic and portfolio standards.

11. Conclusion

The CineScope project successfully delivers a complete, modern movie catalog system with:

- Full CRUD functionality
- TMDB integration
- Manual image upload
- Responsive UI
- Admin dashboard
- Image gallery
- Clean architecture
- Professional documentation

This project demonstrates strong skills in:

- ASP.NET Core MVC
- Entity Framework Core
- API integration
- File handling
- UI/UX design
- Software architecture
- Problem-solving
- Full-stack development